Record - Week 02

Topic: JavaScript ES6 and DOM

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Aim

To build an expense manager using ES6 features, DOM manipulation and the Fetch API with async/await.

Software Requirements

Google Chrome / Microsoft Edge, VS Code, Live Server extension

Procedure

1 Create index.html with the description and amount inputs, an expense list and a total.

2 Write app.js: keep expenses in an array and a render() function that rebuilds the list.

3 Add an event listener on the Add button and on each delete button.

4 Compute the total with reduce() and show it below the list.

5 Add loadFriends(): fetch users from the public API and render them with map().

Program

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Week 02 - JavaScript ES6 &amp;amp; DOM</title>
  <style>
    body { font-family: "Segoe UI", Arial, sans-serif; max-width: 520px; margin: 2rem auto; color: #1f2933; }
    input, button { padding: 0.4rem 0.6rem; font: inherit; }
    li { margin: 0.3rem 0; }
    li button { margin-left: 0.5rem; }
    #total { font-weight: bold; }
  </style>
</head>
<body>
  <h1>Expense Manager</h1>
  <input id="descInput" placeholder="Description">
  <input id="amountInput" type="number" min="1" placeholder="Amount">
  <button id="addBtn">Add</button>
  <ul id="expenseList"></ul>
  <p id="total">Total: 0</p>

  <h2>Friends to Split With (fetched)</h2>
  <button id="loadBtn">Load Friends</button>
  <ul id="friendList"></ul>

  <script src="app.js"></script>
</body>
</html>
```

app.js

```
// ---------- DOM manipulation ----------
const descInput = document.getElementById("descInput");
const amountInput = document.getElementById("amountInput");
const addBtn = document.getElementById("addBtn");
const expenseList = document.getElementById("expenseList");
const totalEl = document.getElementById("total");

let expenses = [];

const render = () => {
  expenseList.innerHTML = "";
  expenses.forEach(({ id, desc, amount }) => {
    const li = document.createElement("li");
    li.textContent = `${desc} - Rs. ${amount}`;

    const del = document.createElement("button");
    del.textContent = "x";
    del.addEventListener("click", () => {
      expenses = expenses.filter((e) => e.id !== id);
      render();
    });

    li.appendChild(del);
    expenseList.appendChild(li);
  });

  const total = expenses.reduce((sum, e) => sum + e.amount, 0);
  totalEl.textContent = `Total: Rs. ${total}`;
};

const addExpense = () => {
  const desc = descInput.value.trim();
  const amount = Number(amountInput.value);
  if (!desc || !(amount > 0)) return;

  expenses.push({ id: Date.now(), desc, amount });
  descInput.value = "";
  amountInput.value = "";
  render();
};

addBtn.addEventListener("click", addExpense);

// ---------- Async / fetch ----------
const loadBtn = document.getElementById("loadBtn");
const friendList = document.getElementById("friendList");

async function loadFriends() {
  try {
    const res = await fetch("https://jsonplaceholder.typicode.com/users");
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const users = await res.json();
    friendList.innerHTML = users
      .map(({ name, email }) => `<li>${name} - ${email}</li>`)
      .join("");
  } catch (err) {
    friendList.innerHTML = `<li>Error: ${err.message}</li>`;
  }
}

loadBtn.addEventListener("click", loadFriends);
```

Output

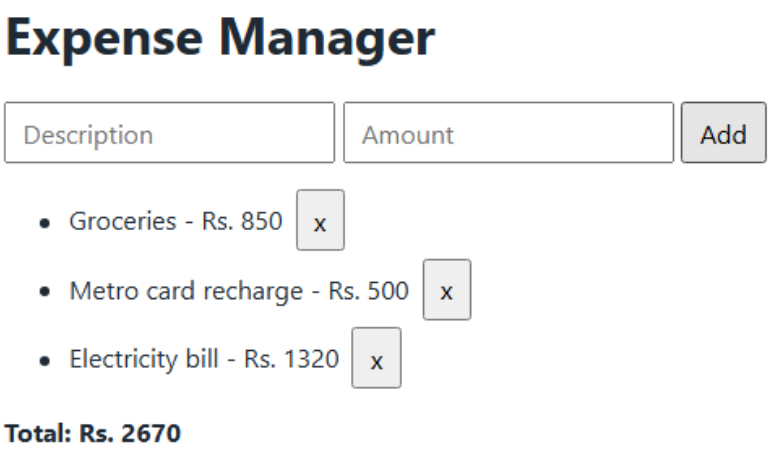

Fig: dom manipulation output

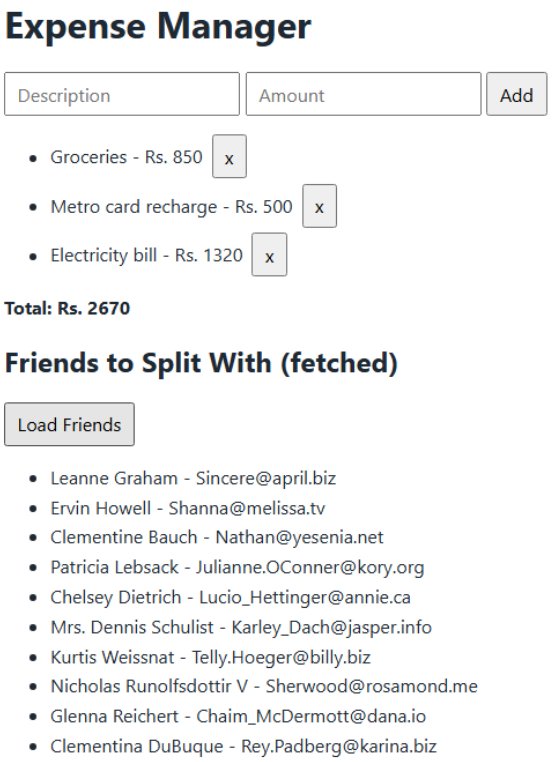

Fig: async fetch output

Result

The program was executed successfully and the expected output was obtained.